

Brazil’s soy land footprint is shifting toward China and toward the agricultural frontier

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Abstract

TODO: Data and Methods working document. The full abstract lives in the main manuscript. Placeholder so the file compiles standalone.

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1 Data

The model is built entirely from publicly and freely available data. Four groups of inputs are combined: (i) Brazilian subnational statistics on soy production, processing, domestic use and trade; (ii) national commodity balances and feed parameters; (iii) administrative and environmental geographies; and (iv) the global input–output backends and the independent validation benchmark. Unless otherwise stated, the series are annual, cover the period 2001–2022 and are broken down by Brazil’s 5,570 municipalities (*municípios*). Where a source exists for only a single reference year, that year is stated and the layer is held constant. Table 1 summarises every input, and all access links are collected in Appendix I. A downloader for the machine-retrievable sources is included with the code.

1.1 Brazilian subnational statistics

Production and population (IBGE): Municipal soybean production and the planted and harvested area data come from the SIDRA system (Table 1612), obtained via its public API. The resident population (Table 6579; 2022 Census) serves as a proxy for household soy oil consumption.

Livestock (IBGE): Municipal herds for ten animal categories (Table 3939) and milked cow counts (Table 94) provide the feed demand base. Feedlot cattle, last reported in the 2006 agricultural census (the 2017 census dropped the confinement question), are carried forward from the 2006 municipal pattern using the national ABIEC/Athenagro confinement series (see Methods).

Processing and biodiesel capacity (ABIOVE, ANP): Soy crushing and refining capacity comes from two manually downloaded ABIOVE products: the consolidated state-level installed capacities and the annual plant roster surveys (*Pesquisa de Capacidade*), whose changing sheet layouts are handled by a per-year dispatcher in the code. Refining and bottling capacities come from the refining sheets of the same products. Years for which surveys are missing from the ABIOVE archive (2000–2002,

2016–2017 and 2021) are replaced with the previous available survey. Biodiesel plant capacity and the regional soy feedstock share are taken from the ANP Statistical Yearbook.

Household consumption and storage (IBGE): State-level per capita soy oil acquisition from the Household Budget Survey (POF 2017–18, the most recent) is used to distribute national food oil use and is held constant across years. Grain storage capacity, which allocates stock changes, is aggregated from the georeferenced warehouse registry underlying the IBGE Stocks Survey (2014 snapshot, held constant).

Municipal trade (COMEX/MDIC): Municipality-level exports and imports are sourced from the COMEX Stat customs database and filtered by the HS4 soy codes 1201 (beans), 1507 (oil) and 2304 (cake).

1.2 National balances and feed parameters (FAO)

Commodity balances: The national supply-use balances against which all municipal totals are reconciled are the FAO Food Balance Sheets for Brazil, items 2555 (soybeans), 2571 (soybean oil) and 2590 (soybean cake), taken from the FAOSTAT bulk download.

Bilateral trade: Re-export disentangling draws on the FAOSTAT Detailed Trade Matrix (items 236, 237 and 238 for beans, oil and cake).

Livestock systems and feed rates: Herds are categorised by production system using the Gridded Livestock of the World rasters (GLW3, reference year 2010) [1] and the GLEAM ruminant system map. They are then converted to soy feed demand using GLEAM per-animal intake rates [2]. These are static layers: the resulting system shares are held constant while herd sizes remain annual (see Methods).

1.3 Geographies

Boundaries: Municipal polygons are IBGE administrative meshes retrieved with the `geobr` package [3]; state outlines for mapping use `GADM` v3.6.

Biomes and frontier: The footprint origin is attributed to biomes using the IBGE 1:250,000 biome layer. The agricultural frontier is the Amazon biome plus MATOPIBA, operationalised as the Cerrado biome within Maranhão, Tocantins, Piauí and Bahia, closely matching the official Embrapa delineation [4].

Grid refinement (optional): A 30m visual refinement uses MapBiomas annual soy tiles [5], exported via Google Earth Engine. This refinement is illustrative only and does not affect the load-bearing capacity of the results.

Transport network: The reported model routes flows based on straight-line distances between municipal centres (see Methods). Four multimodal layers — OpenStreetMap roads, DNIT waterways, ANTT rail lines and cargo, and ANTAQ ports and port cargo — support an experimental cost-based transport variant stored in the repository and used for descriptive analyses. These layers do not affect the reported results.

1.4 Global models and benchmark

FABIO. Subnational flows are incorporated into the Food and Agriculture Biomass Input–Output model [6]. From 2010 onwards the version 2 build is used (2010–2023; 22,263 processes, 181 regions $\times$ 123 items, with 123 environmental extensions). As the v2 build did not exist before 2010, the years 2001–2009 use the corresponding inputs of the v1.1 build: bilateral trade details, land use extensions, feedstock optimisation results and v1 balanced trade and commodity balance sheets. Version bridging and the quantified v1.1/v2 seam are described in the ‘Methods’ section.

EXIOBASE. The non-soy background economy is EXIOBASE 3 [7] in the product-by-product transformation (2000–2022; 49 regions $\times$ 200 products), hybridised with FABIO for the non-food side of the footprint.

Trase (validation only). Modelled origin-to-importer flows are validated against the Trase Brazil soy composite v2.6.1 (2004–2022), which links producing municipalities to importing countries from per-shipment customs and logistics records [8, 9]. No Trase data enter the model itself.

1.5 Coverage and summary

The Footprint series is reported for the period 2001–2022. FABIO v2 covers the period from 2010 onwards, while the FABIO v1.1 build provides the backend for the period from 2001 to 2009. The year 2000 is excluded pending the resolution of a demand-side allocation issue in that year’s build (see Methods). The Trase benchmark supports validation for 2004–2022.

Table 1 Input datasets, by role. All are publicly and freely available; access links are collected in Appendix I. “Munis” = Brazilian municipalities; “exp.” = experimental variant only.

Dataset	Provider	Role	Coverage
Soy production, area (t.1612)	IBGE (SIDRA)	production	munis, annual
Population (t.6579/9514)	IBGE (SIDRA)	food-use proxy	munis, annual
Herds (t.3939), milk cows (t.94)	IBGE (SIDRA)	feed demand	munis, annual
Feedlot cattle (t.919)	IBGE (Census)	feed demand	munis, 2006 pattern
Crush/refining capacity	ABIOVE	processing	munis, annual
Biodiesel capacity	ANP	other (biodiesel) use	munis, annual
Soy-oil acquisition (POF)	IBGE	food-oil allocation	states, 2017–18
Grain storage (warehouse registry)	IBGE	stock allocation	munis, 2014 static
Commodity balance (FBS)	FAO (FAO-STAT)	national reconciliation	Brazil, annual
Gridded livestock (GLW3)	FAO (Data-verse)	system split	10 km, 2010
Production systems, feed (GLEAM)	FAO	feed rates	10 km, static
Municipal trade	COMEX/MDIC	exports/imports	munis, annual
Road network	OSM (Geofabrik)	transport (exp.)	national
Waterways	DNIT	transport (exp.)	national
Rail lines, stations, cargo	ANTT	transport (exp.)	national, 2006–21
Ports, port cargo	ANTAQ	transport (exp.)	national, 2017–22
Municipal boundaries	IBGE (geobr)	geometry	munis
State boundaries	GADM v3.6	mapping	states
Biomes 1:250k	IBGE	origin attribution	national
Soy land tiles	MapBiomass (GEE)	grid refinement	30 m, annual
FABIO v2 (MRIO)	WU Vienna	supply-chain nesting, 2010–2022	global, 2010–23
FABIO v1.1 (MRIO)	WU Vienna	supply-chain nesting, 2001–2010	global, 1986–2013
EXIOBASE 3 pxp	EXIOBASE	tracing soy flows to non-food uses	global, 2000–22
Detailed trade matrix	FAO (FAO-STAT)	re-exports/trade	Brazil, annual
Trase composite v2.6.1	Trase	validation benchmark	munis, 2004–22

1.6 Data availability

All input data are publicly and freely available from the providers listed in Appendix I. They are not redistributed with the code because of differing licences; a download script covers the machine-retrievable sources, and the datasets requiring manual retrieval (ABIOVE, ANP, POF, MapBiomass tiles) are documented step by step in the repository. The intermediate and final model outputs underlying the figures will be deposited on Zenodo upon publication.

1.7 Code availability

The complete modelling pipeline (data preparation, transport allocation, FABIO/EXIOBASE nesting, footprint accounting, validation and figures) is written in R and Python and released under the GNU GPL v3 at <repository URL --- TODO>. The results reported here have no proprietary software dependency. The experimental multimodal-transport module included in the repository (not used for the reported results) additionally requires a GAMS licence.

2 Methods

We couple a subnational (municipality-level) reconstruction of Brazil’s soy supply chain with the global FABIO physical multi-regional input–output (MRIO) framework [6], and attribute the resulting land-use footprint back to municipalities of origin and forward to countries of final consumption, annually for 2001–2022. The construction proceeds in six stages, summarized in Figure 1: (i) municipal soy accounting; (ii) livestock-system disaggregation and feed demand; (iii) reconciliation with national commodity balances; (iv) allocation of physical flows between municipalities by cost-minimizing optimal transport; (v) linkage of origin municipalities to consuming countries, with re-export disentangling; and (vi) nesting into FABIO and land-use footprint accounting. The resulting flows are validated against the independent Trase supply-chain benchmark and a pure downscaling baseline. The subnational tracing framework generalizes the single-year (2013) open model of Trsek [10] to an annual series.

The subsections below describe, for each stage, what is done, with which data, and to what outcome; one subsection per stage, so that Sections 2.1–2.6 correspond to stages 1–6. All equations, data-processing choices and implementation detail are collected in Appendices A–H; a script-level mapping of each stage to the released pipeline code is provided with the repository.

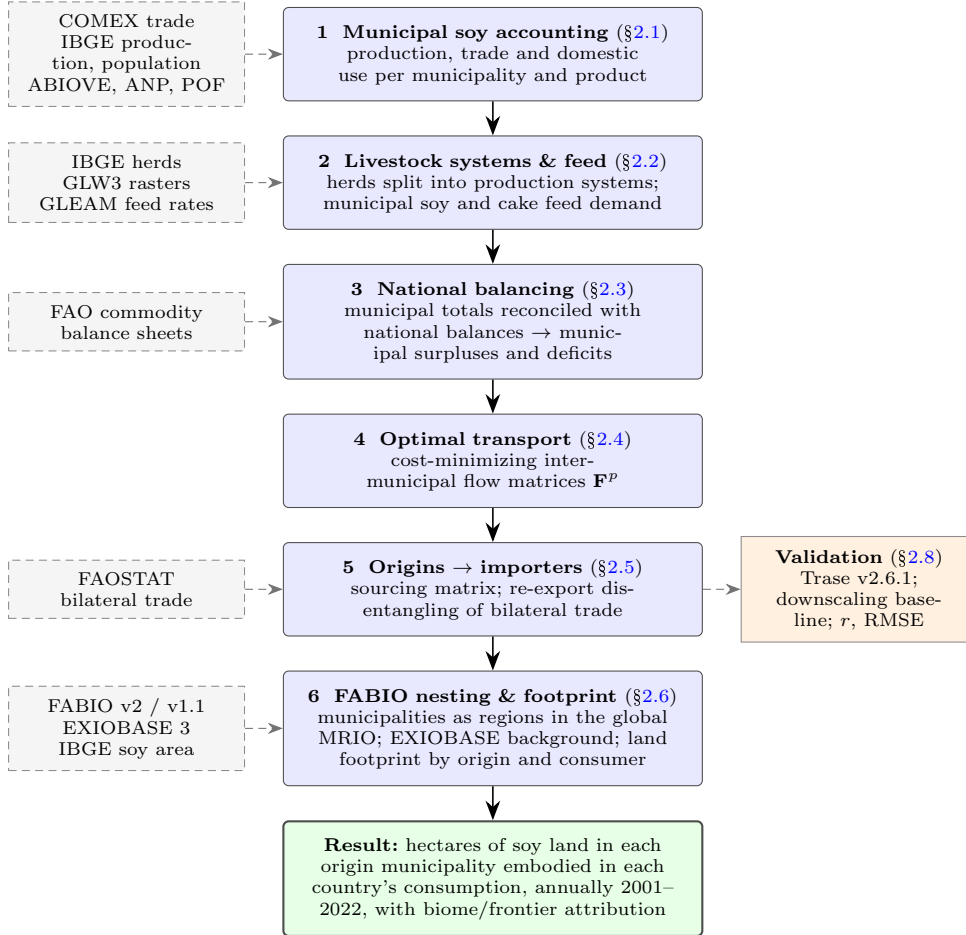


Fig. 1 Workflow of the study. Solid boxes are the six model stages (Sections 2.1–2.6); dashed boxes are the main data inputs at each stage; the orange box is the independent validation of the reconstructed flows (Section 2.8). Stages 1–5 reconstruct Brazil’s subnational soy supply chain; stage 6 embeds it in the global FABIO/EXIOBASE input–output framework and computes the land-use footprint.

2.1 Municipal soy accounting

In the first stage, a complete account of supply, trade and domestic use is compiled for each Brazilian municipality for each of the three soy products: soybeans, soy oil and soy cake (the protein meal remaining after the beans are crushed for oil). Exports and imports are obtained from COMEX customs records at municipality level, production data comes from IBGE municipal surveys, and processing, storage and biodiesel infrastructure data comes from ABIOVE, georeferenced warehouse registers and ANP. Each item of the national FAO commodity balance for domestic use (crush, food, seed, other uses (e.g. biodiesel) and stock changes) is then distributed across municipalities according to the weight most closely tied to that use (see Table 2): installed crushing capacity for the crush, household soy oil consumption for food use, and so on. The outcome is a municipal supply–use account for beans, oil and cake that reproduces the national commodity balance. [The pipeline-specific processing choices and the treatment of stock changes are detailed in Appendix A.](#)

Table 2 Domestic-use allocation: each item is a national commodity-balance total distributed across municipalities in proportion to the stated weight w_m (Appendix A).

Use item	Allocation weight w_m
Processing (crush)	installed crushing capacity
Food (beans, oil)	per-capita soy-oil acquisition $\times$ population
Other (biodiesel)	soy-attributable biodiesel capacity
Seed	soybean production
Stock change	storage capacity (beans), production (oil, cake); addition $\rightarrow$ use, withdrawal $\rightarrow$ supply

2.2 Livestock systems and feed demand

The next step is to estimate the municipal feed demand for soybeans and soy cake. Since the amount of soy consumed by each animal varies greatly depending on the livestock production system, the IBGE’s herd counts at municipal level are first divided into production systems (extensive, intensive, and industrial monogastrics; grassland- and mixed-based ruminants; dairy and meat) using the 2010 FAO gridded livestock maps (GLW3 and GLEAM). The number of animals in each system is then multiplied by the amount of soy and soy cake consumed by each animal in that system, as determined by GLEAM. The bottom-up totals are then rescaled to the national feed use reported by the FAO (see Figure 2). The outcome is the municipal feed demand for beans and cake by species and production system, which is consistent with the national balance. [The splitting rules and the implications of keeping the 2010 system shares fixed throughout the study period are provided in Appendix B.](#)

2.3 National balancing

Next, the municipal accounts are reconciled with the national supply–use identity of each product (production + imports + stock withdrawal = exports + food + feed + seed + processing + other). Municipal trade is harmonised with the FABIO/FAOSTAT bilateral trade classification. Any remaining national residual is absorbed into the stock term and each balance item is proportionally rescaled so that every municipal aggregate matches its national total exactly. The outcome is a balanced set of municipal net positions for each product (a surplus where local supply exceeds use, a deficit where use exceeds supply). The supply–use identity, the surplus/deficit decomposition, and the treatment of net stock withdrawals are explained in Appendix C.

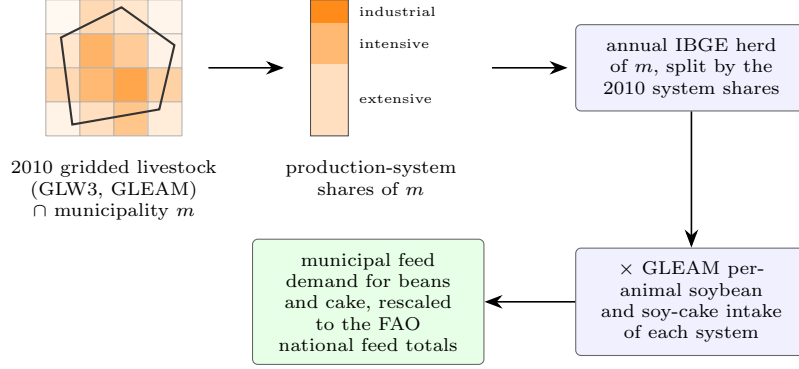


Fig. 2 From herds to municipal soy feed demand. Municipal herd counts (annual) are divided into production systems using the shares implied by the 2010 gridded livestock rasters within each municipality. Each system’s herd is multiplied by its per-animal soy and soy-cake intake, and the totals are then scaled to match the national feed use reported by the FAO.

2.4 Subnational transport allocation

In this step, municipal surpluses and deficits are connected via optimal transport that minimises costs. For each product, the inter-municipal flow matrix $\mathbf{F}^p$ is the cheapest way to assign surpluses to deficits while conserving mass. The premise is that bulk-commodity logistics minimise freight costs, which increase with distance. Three allocation variants are analysed in order of increasing supply-chain structure: a pure downscaling baseline, which ignores the domestic network and splits each destination’s imports across municipalities according to their production shares; the Euclidean model, which solves the transportation problem using straight-line distances between municipal centres; and a multimodal variant, which uses Brazil’s actual road, rail and waterway networks for the same flows, taking into account mode-specific freight costs and corridor capacities (all three variants are specified in Appendix D). When compared with the independent Trase benchmark, the three models produce similar results at the aggregate level: the pooled municipality×destination correlation averages 0.71, 0.69 and 0.69 respectively over the period from 2010 to 2022, but differences emerge at the level of individual destinations (see Section 2.8). The reported results use the Euclidean model: the multimodal variant does not reproduce the benchmark better than the Euclidean model in any year, and requires network inputs that are unavailable for parts of the study period. The downscaling baseline produces no inter-municipal flows, which are required for the tracing of processing, feed and domestic consumption in stage 5. The outcome is a deterministic inter-municipal flow matrix per product whose margins reproduce the municipal surpluses and deficits by construction. [The formal problem statement and the solvers are specified in Appendix D](#); [the multimodal benchmark results are reported in Section 2.8](#).

2.5 From origins to consumers

The COMEX customs records of stage 1 attach two labels to every exported tonne: the municipality that shipped it and the country that first imported it. However, neither label contains the information that the footprint needs: the municipality that grew the soy and the country that finally consumes it. This stage corrects both issues. Within Brazil, the transport flows of stage 4 serve as the tracing device. Completed with each municipality’s retained supply, they yield sourcing shares that trace every exported (and domestically consumed) tonne back to its origin municipality (Figure 3). Oil and cake are traced one step further back to the municipalities where the beans were grown. Moving forwards across borders, exported tonnes often pass through intermediate countries before being consumed. The FAOSTAT bilateral trade data are therefore corrected for re-exports [11]: each tonne is followed through the trade network until it reaches the country where it is finally consumed (Figure 3). In this calculation, Brazil enters as its individual municipalities rather than as a single country. The outcome is

an origin-to-consumption matrix linking every producing municipality to every consuming country. The equations, the underlying proportional-mixing assumption and its numerical implementation are given in Appendix E.

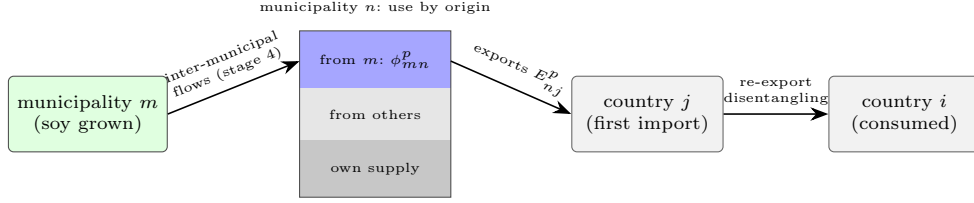


Fig. 3 Tracing a tonne from origin to final consumer. The soy grown in municipality m reaches municipality n through the inter-municipal flows of stage 4, so the share ϕ_{mn}^p of n 's use is attributed to m (highlighted). The observed exports E_{nj}^p of n carry that share to the country of first import j , and the re-export disentangling reattributes it to the country of final consumption i . The corresponding equations are given in Appendix E.

2.6 Nesting into FABIO and footprint accounting

In the final stage, the reconstructed municipal soy economy is embedded in the FABIO physical MRIO, in which a process is a region $\times$ commodity unit. Brazil's three national soy commodities and processes are removed and replaced by their $\sim 16,700$ municipal counterparts, so that municipalities enter the global model as additional regions, each carrying its own production, soy area, feed use and origin-resolved trade from the previous stages (Figure 4). The regional accounts are then linked through the re-export-corrected trade of stage 5 into a single global input-output model, following the published FABIO procedure [6]. Soy used outside the food system, for example as biodiesel, is traced onwards by coupling FABIO to the EXIOBASE model of the wider economy [7]. From 2010 onwards the model runs on the FABIO v2 backend; for 2001–2009 it switches to inputs from the FABIO v1.1 build, and the seam between the two backends is quantified on the overlap years 2010–2013. The backend bridging and seam analysis are documented in Appendix F; the assembly of the nested model and the EXIOBASE coupling follow the published FABIO procedures [6, 7].

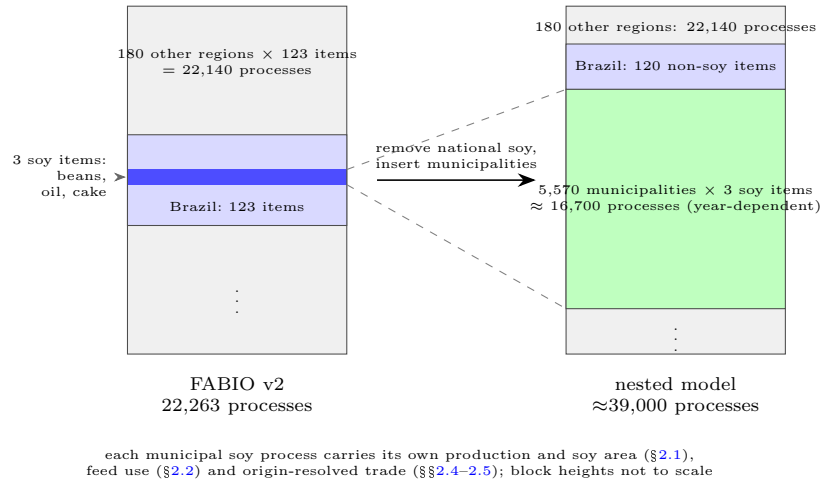


Fig. 4 Nesting municipal soy into the global input-output system: the three national Brazilian soy commodities are removed from the process list and replaced by one soy process set per municipality, populated with the municipal accounts of stages 1–5; all other regions, and Brazil's remaining commodities, are untouched. The assembly follows the published FABIO procedure [6]; the backend coverage and the seam analysis are detailed in Appendix F.

Land-use footprint accounting.

The direct land use of each process is its harvested soy area, taken from IBGE for the municipal soy processes and from the FABIO land-use extension for all others. Dividing by output gives a land intensity per unit of product, which the footprint calculation traces through the supply chains to each country’s final demand, food and non-food uses alike. Each municipality is assigned to a biome (IBGE 1:250,000 layer), and the agricultural frontier is defined as the Amazon biome plus MATOPIBA. The result is hectares of soy land in each origin municipality embodied in each country’s consumption, per year, with biome and frontier attribution. The treatment of stocks, footprints by consumed product, and the optional 30 m grid refinement are described in Appendix [G](#).

2.7 Headline results

Four outputs illustrate what the pipeline delivers. The footprint grew from 9.4 Mha in 2001 to 41.0 Mha in 2022, with its origin shifting from the southern states to the Cerrado frontier (see Fig. [5](#)). During the same period, the EU-27 and China exchanged roles as the dominant consumer, with this change occurring in 2009–2010 (Fig. [6](#)). Most of the footprint reaches consumers in commodity form. From 2010 onwards, around a quarter is embodied in animal products (Fig. [7](#)). At the finest level, the footprint can be displayed as probability maps per destination on 30 m MapBiomass soy pixels (Fig. [8](#)).

Brazilian soy land-use footprint by destination, municipal origin, 2001 vs 2022 Hectares of soy land embodied in consumption (F_{mass}). All panels share one colour scale; sqrt-scaled.

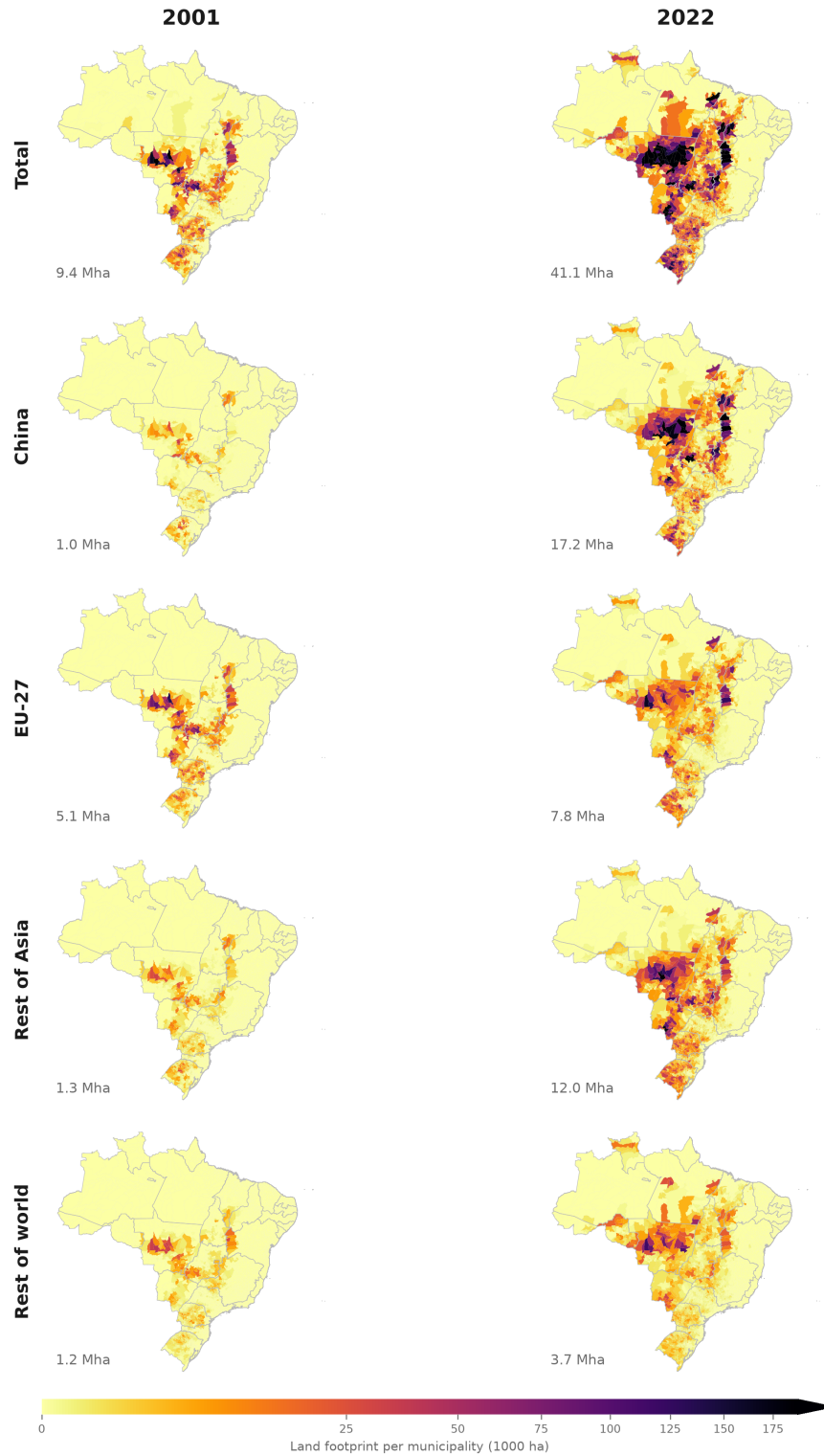


Fig. 5 Municipal origin of the Brazilian soy land-use footprint, 2001 versus 2022, for the total and the four major destination groups; all panels share one square-root colour scale. An interactive year-by-year version is available at <https://lizaburiak.github.io/soyprint-footprint-map/>.

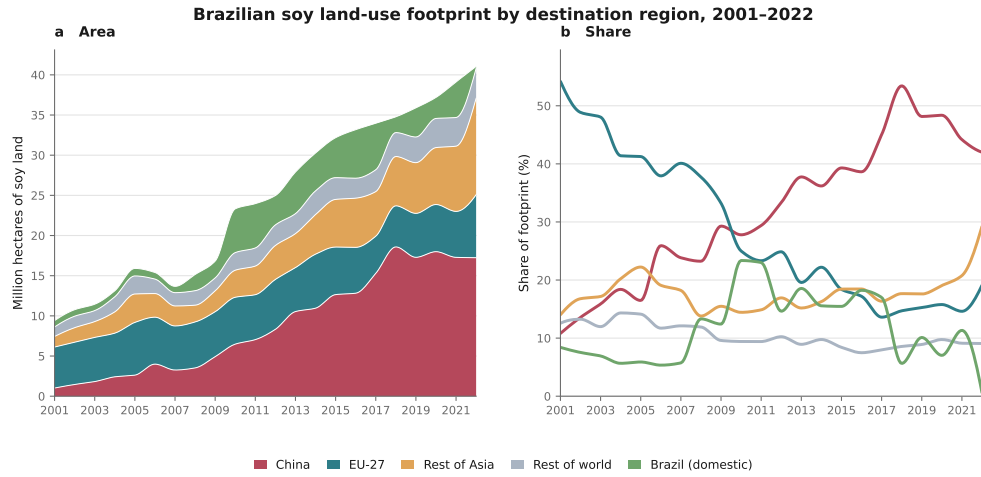


Fig. 6 Footprint by destination region, 2001–2022: (a) absolute footprint, (b) regional shares.

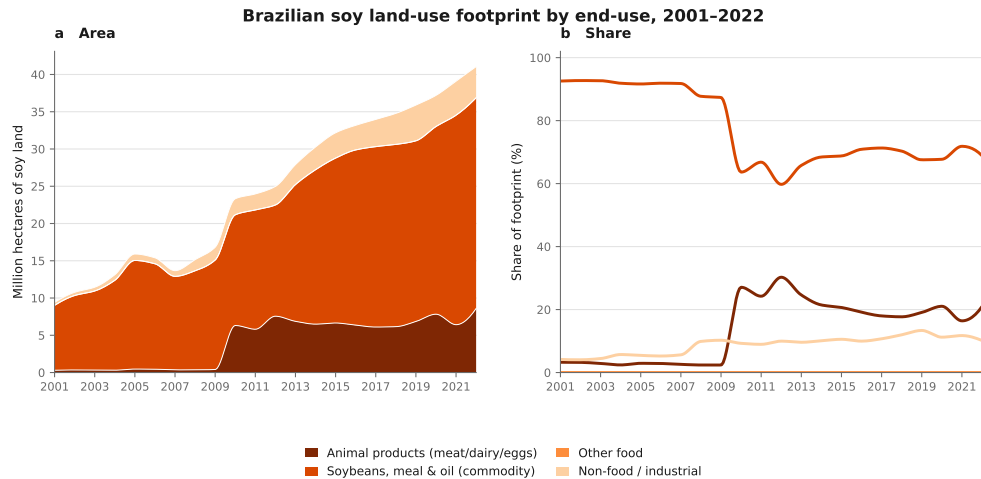


Fig. 7 Footprint by end-use, 2001–2022: (a) absolute footprint, (b) end-use shares.

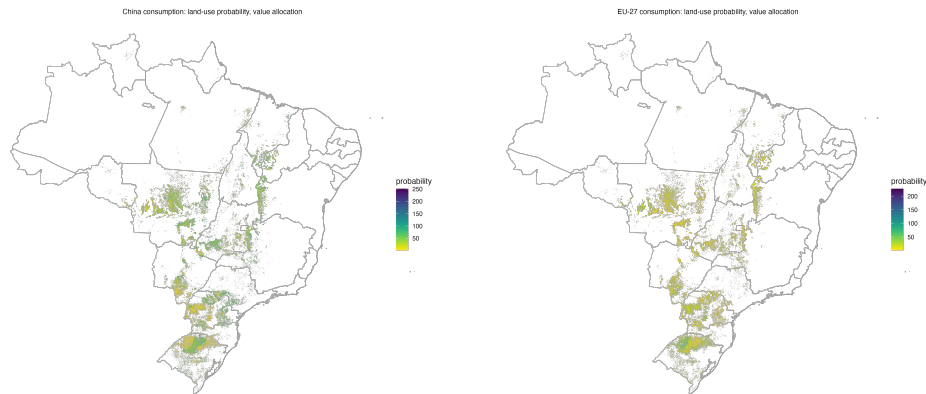


Fig. 8 Probability maps of the 2022 footprint on 30 m MapBiomas soy pixels, for China (left) and the EU-27 (right); colour gives the pixel's probability of serving the destination's consumption (value allocation).

2.8 Validation and uncertainty

The reconstructed origin-to-importer flows are evaluated against the independent Trase Brazil–soy dataset (v2.6.1), which links municipalities of production to countries of first import using per-shipment customs and logistics records [8, 9]. All three allocation variants of Section 2.4 are compared against this benchmark on the municipality×destination flow matrix: the downscaling baseline, the Euclidean model and the multimodal variant (Figure 9). Pooled over all flows the three perform comparably, with mean correlations of 0.71, 0.69 and 0.69 over 2010–2022. The value added by the supply-chain structure shows at the destination level: the mean per-destination correlation is 0.41 for the Euclidean model and 0.42 for the multimodal variant, against 0.28 for downscaling. [Sample construction is given in Appendix H.](#)

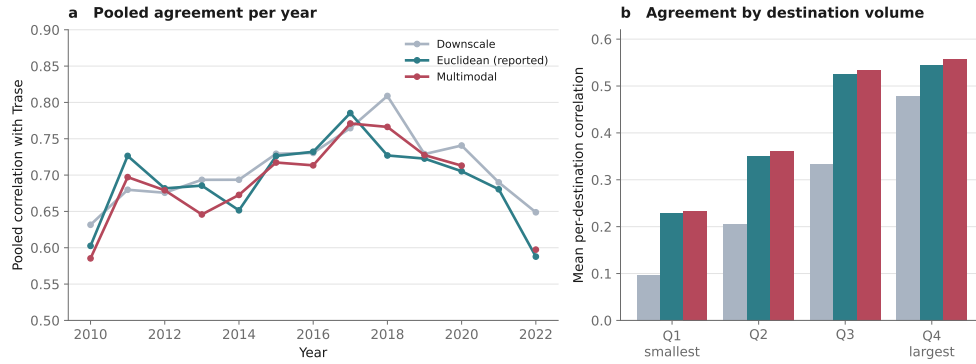


Fig. 9 The three allocation variants against the Trase benchmark, 2010–2022. (a) Pooled municipality×destination correlation per year (the multimodal variant has no 2021 solve). (b) Mean per-destination correlation by import-volume quartile: the supply-chain models separate from downscaling at the destination level, and all variants improve with volume.

Agreement rises with flow size (Figure 10). Pooled over the [91,873](#) matched municipality×destination×year flows of [2010–2022](#), the log-flow correlation with Trase is 0.68 and 54% of the flows fall within a factor of three of the benchmark value. In the upper half of the volume distribution (flows above 1.2 kt, which carry 98% of the benchmark tonnage) the within-factor-three share rises to [57%](#) and the median deviation falls from a factor of [2.6](#) to [2.3](#); per destination, the mean correlation rises from 0.29 in the lower half of import volumes to 0.53 in the upper half. The tonnage that carries the footprint is therefore also the best-validated part of the flow distribution, and below roughly 10^3 t per destination subnational modelling adds no information over downscaling (Appendix [H](#)).

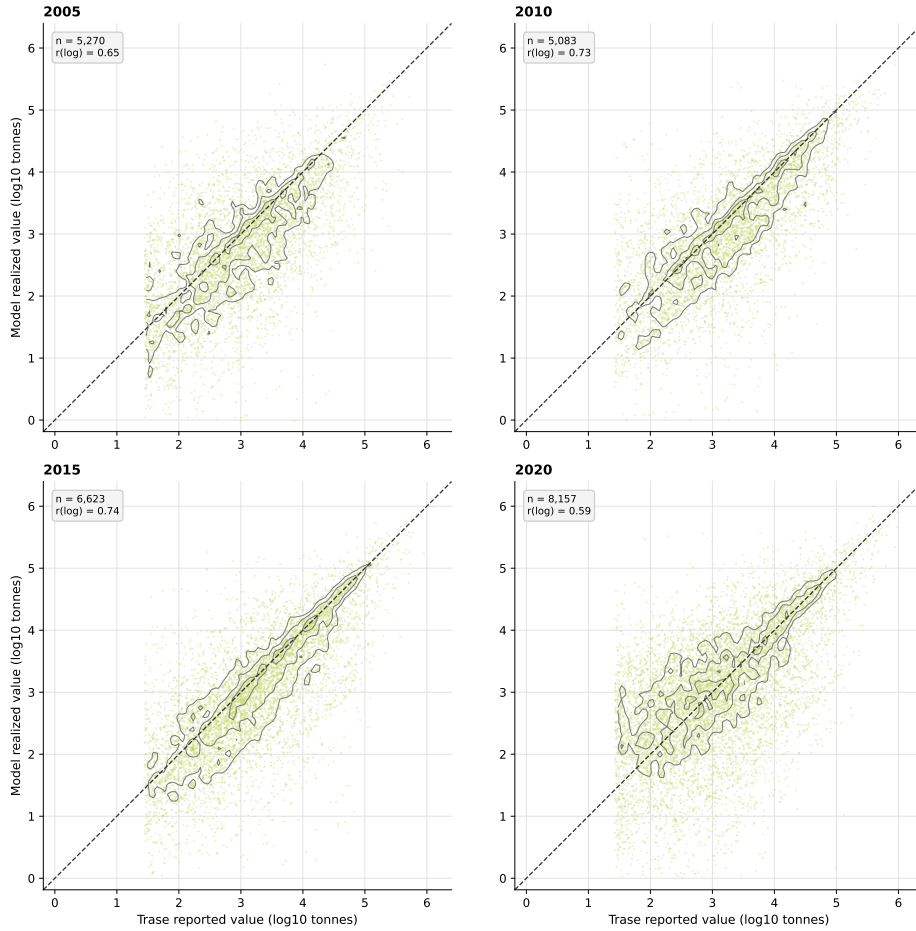


Fig. 10 Agreement with the benchmark grows with flow size (“rocket plots”). Every modelled municipality×destination soy export flow is plotted against the corresponding Trase flow (both $\log_{10}$ tonnes; flows below 1 t on either side are excluded) for four sample years, with density contours and the 1:1 line (dashed); each panel reports the number of matched flows, the log-scale Pearson correlation and the share of flows agreeing within a factor of three. Large flows converge onto the 1:1 diagonal (the “nose”), while small flows scatter (the “plume”); agreement rises systematically with volume, which motivates the volume floor of the decision rule (Appendix H): the tonnage that carries the footprint is also the best-validated part of the flow distribution.

A second, temporal check concerns the two FABIO backends. Before 2010 the pipeline runs on FABIO v1.1 inputs instead of the v2 build (Section 2.6); in the clean overlap years the two backends differ by 6–8% in the total footprint with cross-country correlations above 0.988, so pre-2010 levels are reported unadjusted (Appendix F).

A sensitivity screen perturbs the two input assumptions that the data pin down least: the within-state location of crushing capacity and the feed and livestock parametrisation. Twenty-four perturbed variants of the flow reconstruction are re-run for a representative year (2019) and scored against Trase (Euclidean model, base $r = 0.72$). Crush location accounts for nearly all of the induced variation (random within-state capacity splits span correlations of 0.67 to 0.74), while the feed-rate and herd-composition variants move the correlation by less than 0.01; the perturbation recipes are given in Appendix H.

2.9 Software

The pipeline is implemented in R (4.5; `data.table`, `dplyr`, `Matrix`, `sf`, `transport`) with figures and web-map tooling in Python (3.14; `pandas`, `matplotlib`, `geopandas`). The pipeline, including the multimodal transport variant of Appendix D (Pyomo with the open-source HiGHS solver), has no proprietary software dependency.

3 Data description

3.1 Data records

The complete model output is released as a relational database of municipality-level records, deposited on Zenodo upon publication (DOI: [TODO](#)). So that no particular software stack is required, the same content is provided in three redundant layers: (i) the canonical layer in Apache Parquet, one folder per table, partitioned by year (`<table>/year=YYYY/data.parquet`; 0.6 GB in total), readable directly from R, Python, QGIS and any SQL engine; (ii) a single self-contained DuckDB file (`soyprint.duckdb`, 0.9 GB) with all tables materialised, ready-made analysis views and the build metadata; and (iii) plain CSV (1.2 GB), one file per table, with the three large footprint tables split into one compressed file per year, plus six small summary tables (footprint by country, by end use, by biome and by state, the national soy balance, and the top footprint municipalities per year) that open directly in a spreadsheet.

Figures [11](#) and [12](#) show the schema of the results and input tables respectively; Table [3](#) lists every table with its grain, contents and size. The records fall into three groups. Eight fact tables hold the stage outputs: the municipal soy accounts (`production`, `domestic_use`, `trade`), the transport-model outputs (`transport_flows` and `export_attribution`, each reported for both the Euclidean baseline and the multimodal variant) and the footprint results (`footprint_country`, `footprint_product`, `footprint_animal_country`). Six input tables carry the harmonised inputs exactly as the accounting stages consume them, before any balancing, so that the harmonisation and balancing choices of the 'Methods' section can be re-examined, and the municipal accounting stage re-run, without retrieving the raw sources of Table [1](#); re-running the transport and footprint stages additionally requires the geographic network layers and the FABIO and EXIOBASE backends listed there. Four dimension tables provide the join keys: municipalities are keyed by their 7-digit IBGE code (with name, state, biome, MATOPIBA membership and coordinates), countries by ISO3 code (with continent, region and EU-27 membership, including the five EXIOBASE rest-of-world regions), commodities by FABIO commodity code, and the three soy products (bean, oil, cake) by their HS4 and FABIO items. All quantities are expressed in tonnes, hectares and current USD. The fact tables cover 2000–2022 and the input tables 2000–2023; the footprint records for 2000 are included in the deposit but inherit the backend issue noted in 'Coverage and summary' and are excluded from all reported results.

Two properties of the footprint tables matter for reuse. `footprint_country` and `footprint_product` are complete marginals of the same footprint: each sums to the total footprint of a year, which is verified at build time. `footprint_animal_country` is the only joint country $\times$ product cut the pipeline produces; it covers the eight individual FABIO animal commodities only (roughly half of the total area) and is therefore not a complete decomposition. Provenance is recorded alongside the data: `meta_provenance` lists the source file and its modification time behind every table-year export, and `meta_build` the build timestamp, code version (git commit) and DuckDB version. The DuckDB layer additionally ships four analysis views for the most common cuts: `v_soy_balance` (area, production, processing, exports and domestic use side by side per municipality and year), `v_footprint_by_country`, `v_footprint_by_enduse` and `v_footprint_by_biome`.

Table 3 Data records: tables of the released database. Every fact and input table carries a **year** column; “mun” = municipality (7-digit IBGE code), “prod.” = soy product (bean, oil, cake), “method” = transport variant (Euclidean / multimodal). Rows are totals over all years.

Table	One record per	Values	Years	Rows
Fact tables				
production	mun	planted/harvested area, bean output, crush, oil and cake output	2000–22	128 k
domestic_use	mun × prod. × use category	tonnes (food, feed, seed, processing, other, stock change)	2000–22	545 k
trade	mun × prod. × flow × partner	tonnes, USD (balanced customs records)	2000–22	45 k
export_attribution	mun × prod. × destination × method	tonnes, attributed to the producing mun	2000–22	1.9 M
transport_flows	origin mun × destination	tonnes	2000–22	787 k
footprint_country	mun × prod. × method mun × consumer country × demand type	hectares	2000–22	10.5 M
footprint_product	mun × consumer product	hectares	2000–22	9.1 M
footprint_animal_country	mun × country × animal commodity	hectares	2000–22	47.0 M
Input tables				
input_municipality	mun	area, production, trade, capacities, population, herds, storage	2000–23	133 k
input_trade	mun × prod. × flow × partner	tonnes, USD (raw customs records)	2000–23	49 k
input_cbs_national	prod.	FAO commodity balance, Brazil	2000–23	72
input_livestock_systems	mun × production system	heads (26 categories)	2000–23	2.3 M
input_feed	mun × animal category	bean and cake feed, tonnes	2000–22	1.3 M
input_faostat_trade	prod. × flow × partner	tonnes, USD (FAO-STAT benchmark)	2013–24	2.3 k
Dimension and metadata tables				
dim_municipality	mun	name, state, biome, MATOPIBA, lon/lat	static	5,572
dim_country	country	name, continent, region, EU-27	static	267
dim_commodity	FABIO commodity	item, group, end-use bucket	static	140
dim_soy_product	soy product	HS4 and FABIO codes	static	3
meta_provenance	table × year	source file, modification time	–	305
meta_build	build	timestamp, git commit, versions	–	1

SOYPRINT database (soyprint.duckdb). Fact tables cover 2000–2022 and are hive-partitioned by year in the Parquet layer.

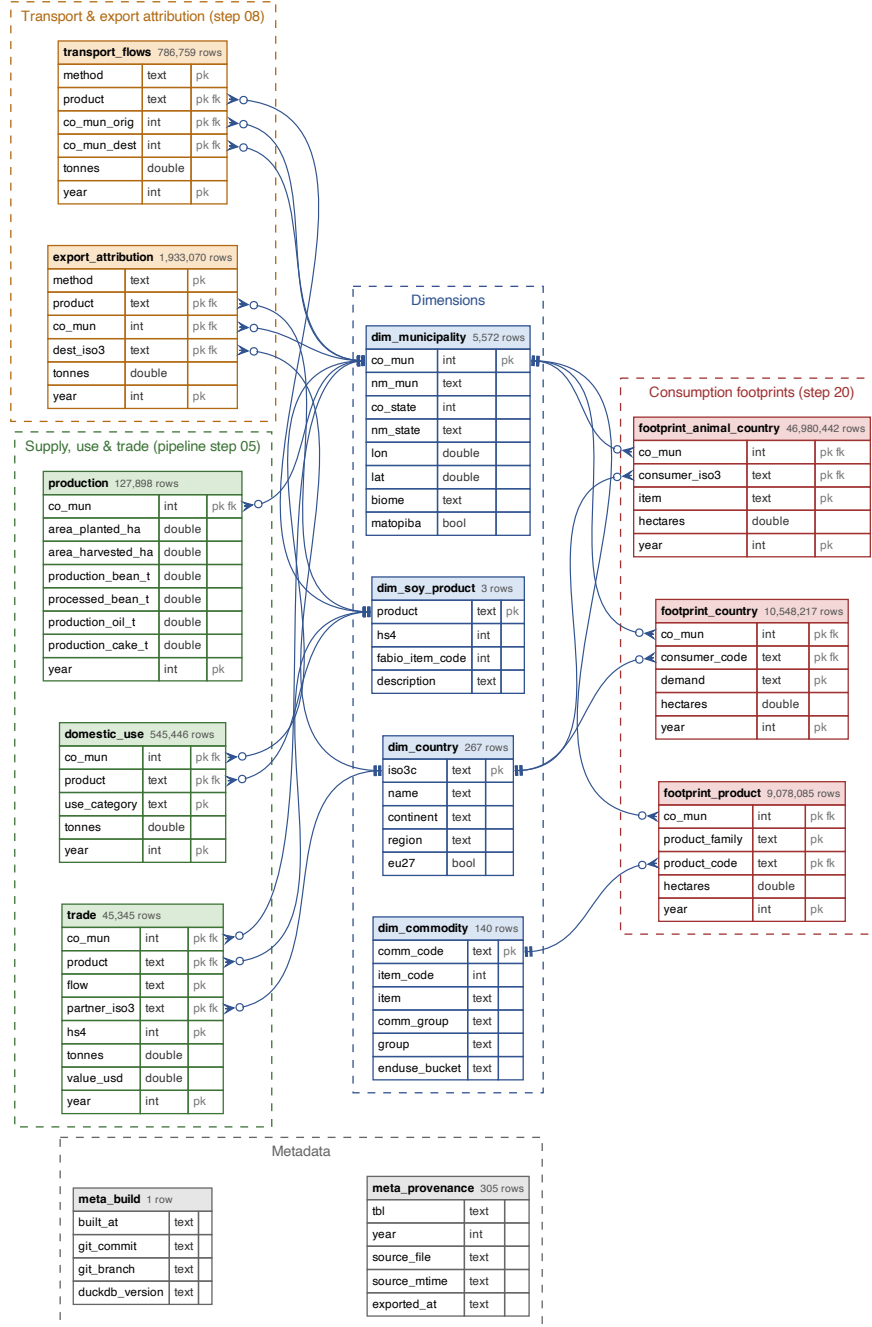


Fig. 11 Schema of the released results database: eight fact tables, grouped by the pipeline stage that produces them, linked to four dimension tables, with build and provenance metadata alongside. Each field is listed with its type, and primary and foreign keys are marked (pk, fk). Fact tables are hive-partitioned by year in the Parquet layer.

SOYPRINT input database (soyprint.duckdb, input_* tables). Cleaned inputs of pipeline steps 00–03, 2000–2023, hive-partitioned by year in the Parquet layer.

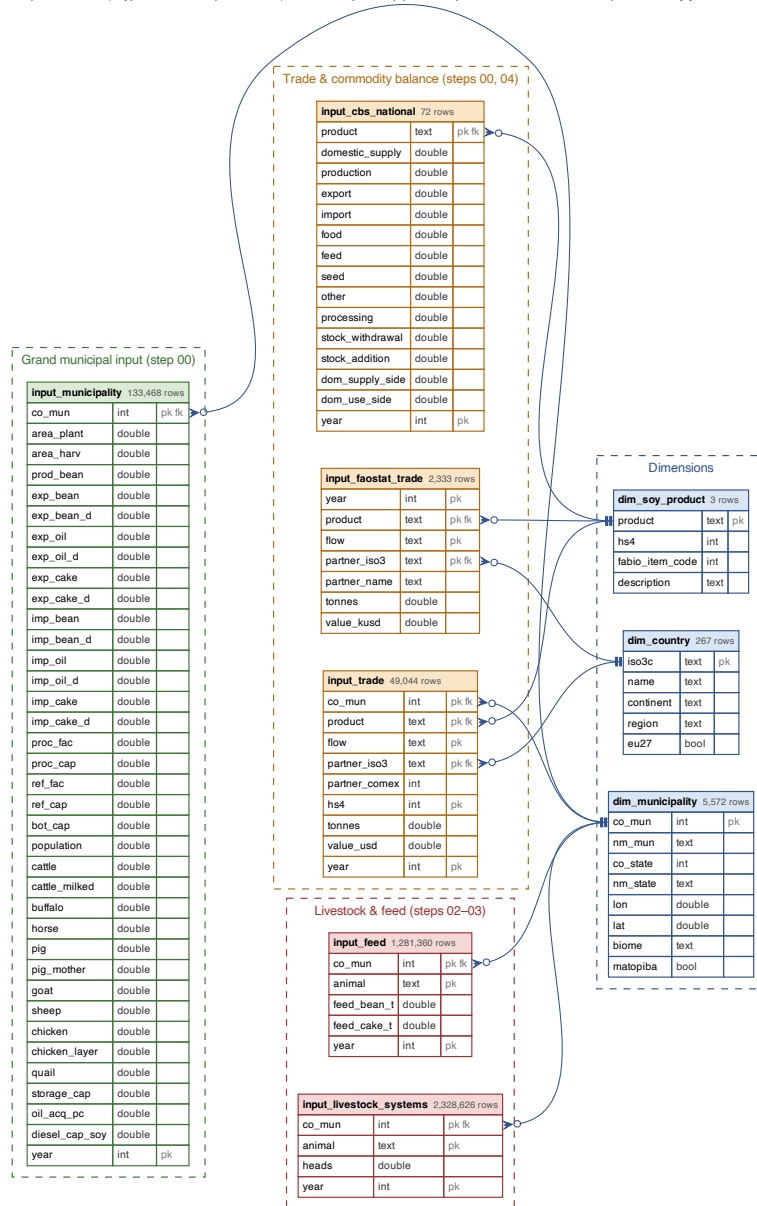


Fig. 12 Schema of the input tables of the released database: the harmonised inputs as the accounting stages consume them (the grand municipal input, the raw customs records, the national commodity balance, the FAOSTAT bilateral benchmark, and the livestock-system and feed tables), linked to the same dimension tables as the results. Each field is listed with its type, and primary and foreign keys are marked (pk, fk). Like the fact tables, the input tables are hive-partitioned by year in the Parquet layer.

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